Reminders

Upcoming due dates
Nothing due this week

Discussion section is a tour of Datahub and an ungraded Python review notebook

Version Control 1.0

Data Science in Practice

Jason G. Fleischer, PhD Dept. of Cognitive Science UC San Diego https://jgfleischer.com

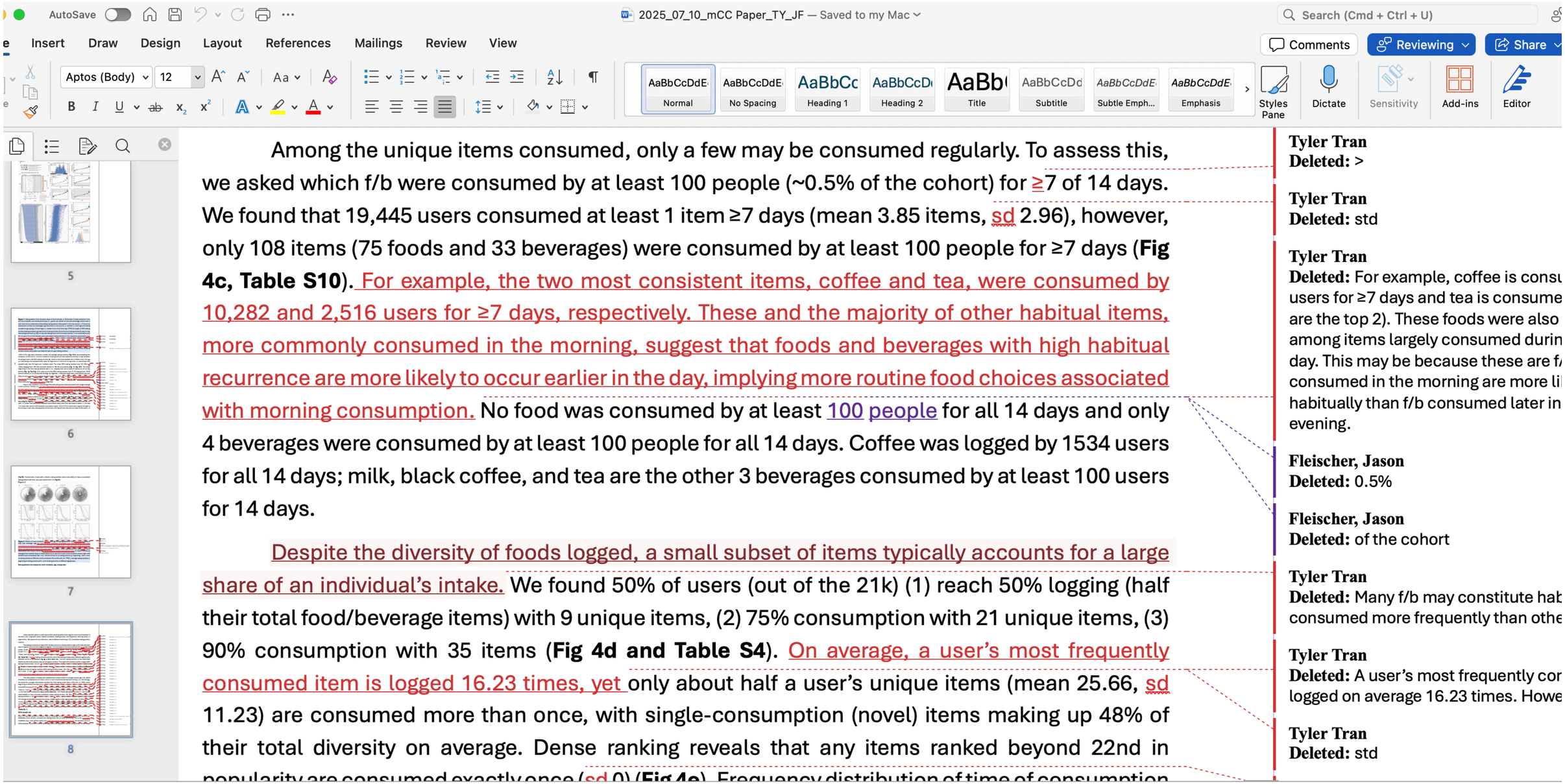
This sucks

archived version of my Documents folder from ~ 2012

- LNAI_fulltext.pdf
- LNAI_v4520.pdf
- Neuromorphic BBD book
- Neuromorphic BBD book - from Desktop
- Neuromorphic BBD book -- FINAL version 99% sure
- neuromorphic book abstract final.pages
- neuromorphic robots book abstract v1.pages
- NeuromorphicBookChapter2011.pdf
- neuroreport_v18_n17_2007.pdf
- pnas_v104_n9_pp3556-3561.pdf
- robotics and automation magazine (conflict at 2012-07-28_00-23-24)
- Robotics and Automation Magazine 2009 final version
- Robotics and Automation Magazine 2009 f***ed up copy due to sync with laptop i think
- Robotics and Automation Magazine 2009 may be jacked tex file
- Robotics and Automation Magazine 2009 not final version, too many refs
- Robotics and Automation Magazine 2009 not the final, too many refs to be it
- Robotics and Automation Magazine 2009.zip
- robotics and automation magazine.pages
- Rome JIN Submission
- SegwaySoccerICRA2006.pdf

Several months after finishing a writing project, I wanted to keep only the final version of the many different revisions… figuring out which one was the version actually sent to the publisher was hard!

A step in the right direction



Among the unique items consumed, only a few may be consumed regularly. To assess this, we asked which f/b were consumed by at least 100 people (~0.5% of the cohort) for ≥7 of 14 days. We found that 19,445 users consumed at least 1 item ≥7 days (mean 3.85 items, sd 2.96), however, only 108 items (75 foods and 33 beverages) were consumed by at least 100 people for ≥7 days (**Fig 4c, Table S10**). For example, the two most consistent items, coffee and tea, were consumed by 10,282 and 2,516 users for ≥7 days, respectively. These and the majority of other habitual items, more commonly consumed in the morning, suggest that foods and beverages with high habitual recurrence are more likely to occur earlier in the day, implying more routine food choices associated with morning consumption. No food was consumed by at least 100 people for all 14 days and only 4 beverages were consumed by at least 100 people for all 14 days. Coffee was logged by 1534 users for all 14 days; milk, black coffee, and tea are the other 3 beverages consumed by at least 100 users for 14 days.

Despite the diversity of foods logged, a small subset of items typically accounts for a large share of an individual's intake. We found 50% of users (out of the 21k) (1) reach 50% logging (half their total food/beverage items) with 9 unique items, (2) 75% consumption with 21 unique items, (3) 90% consumption with 35 items (**Fig 4d and Table S4**). On average, a user's most frequently consumed item is logged 16.23 times, yet only about half a user's unique items (mean 25.66, sd 11.23) are consumed more than once, with single-consumption (novel) items making up 48% of their total diversity on average. Dense ranking reveals that any items ranked beyond 22nd in popularity are consumed exactly once (sd 0) (**Fig 4e**). Frequency distribution of time of consumption

Tyler Tran
Deleted: >

Tyler Tran
Deleted: std

Tyler Tran
Deleted: For example, coffee is consu users for ≥7 days and tea is consume are the top 2). These foods were also among items largely consumed durin day. This may be because these are f/ consumed in the morning are more li habitually than f/b consumed later in evening.

Fleischer, Jason
Deleted: 0.5%

Fleischer, Jason
Deleted: of the cohort

Tyler Tran
Deleted: Many f/b may constitute hab consumed more frequently than othe

Tyler Tran
Deleted: A user's most frequently cor logged on average 16.23 times. Howe

Tyler Tran
Deleted: std

Version Control

- Enables multiple people to simultaneously work on a single project.

- Each person edits their own copy of the files and chooses when to share those changes with the rest of the team.

- Thus, temporary or partial edits by one person do not interfere with another person's work

What is version control?

A way to manage the evolution of a set of files

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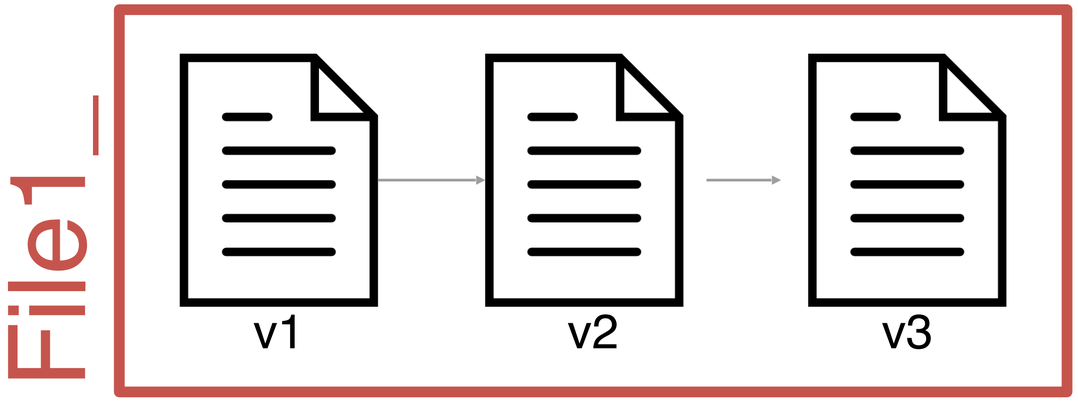
A way to manage the evolution of a set of files



What is version control?

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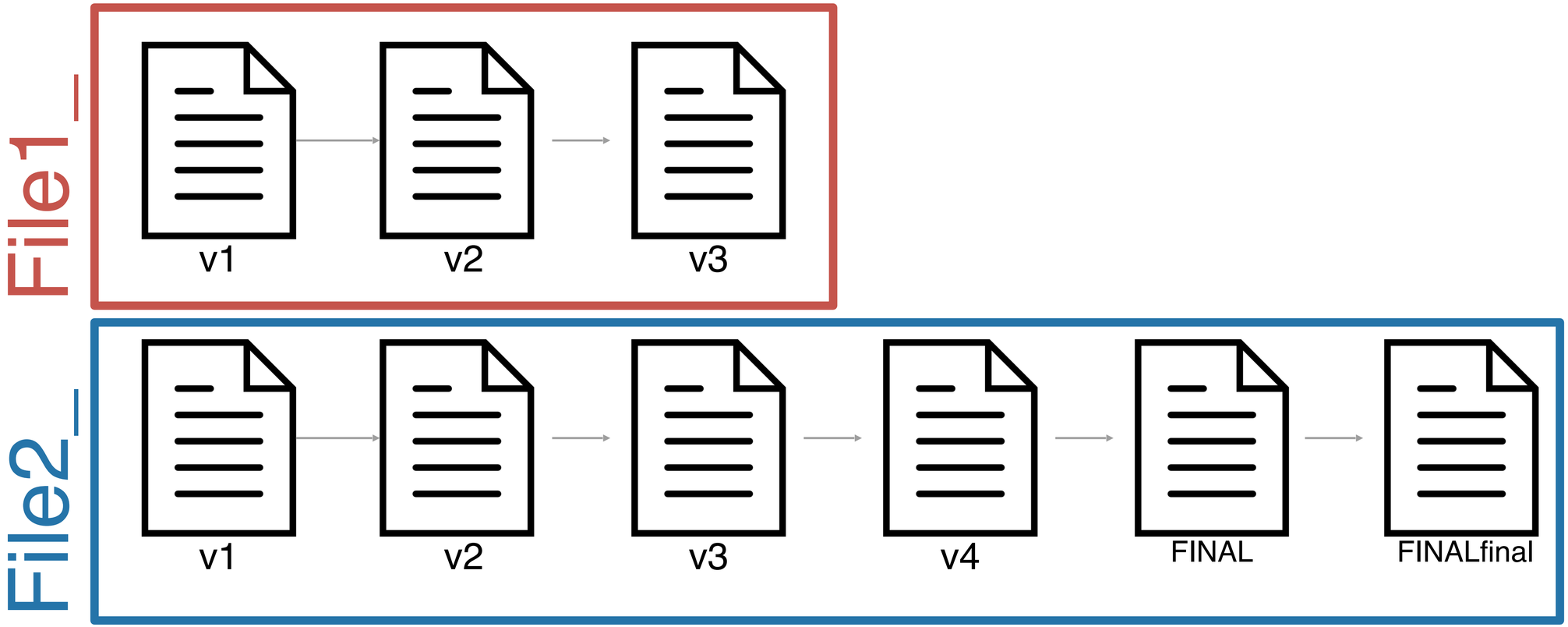

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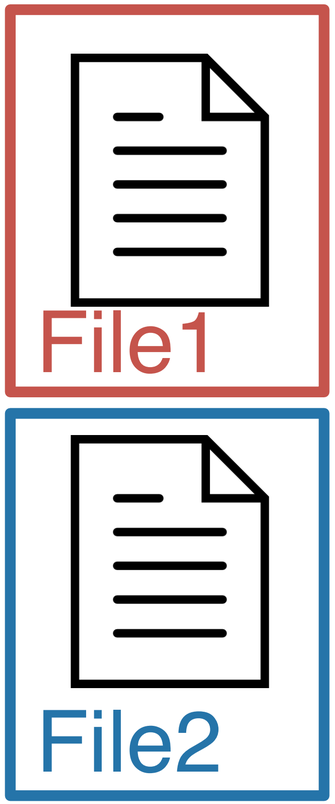
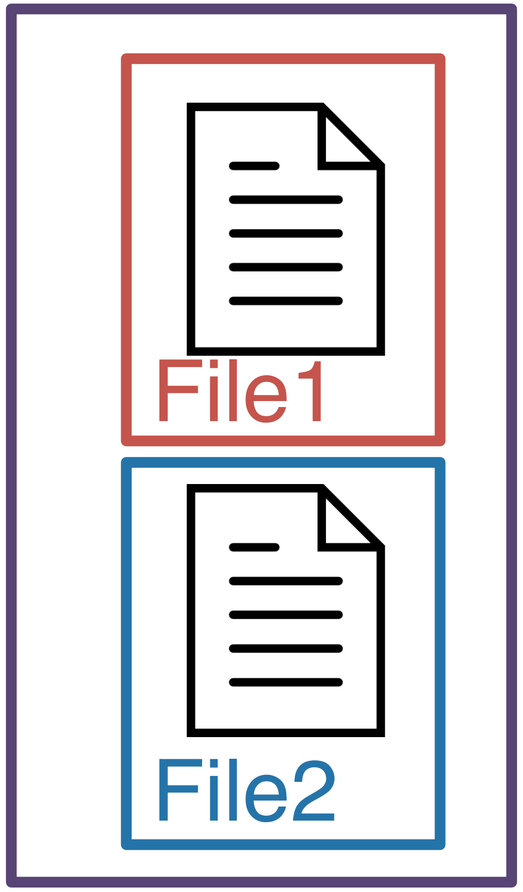


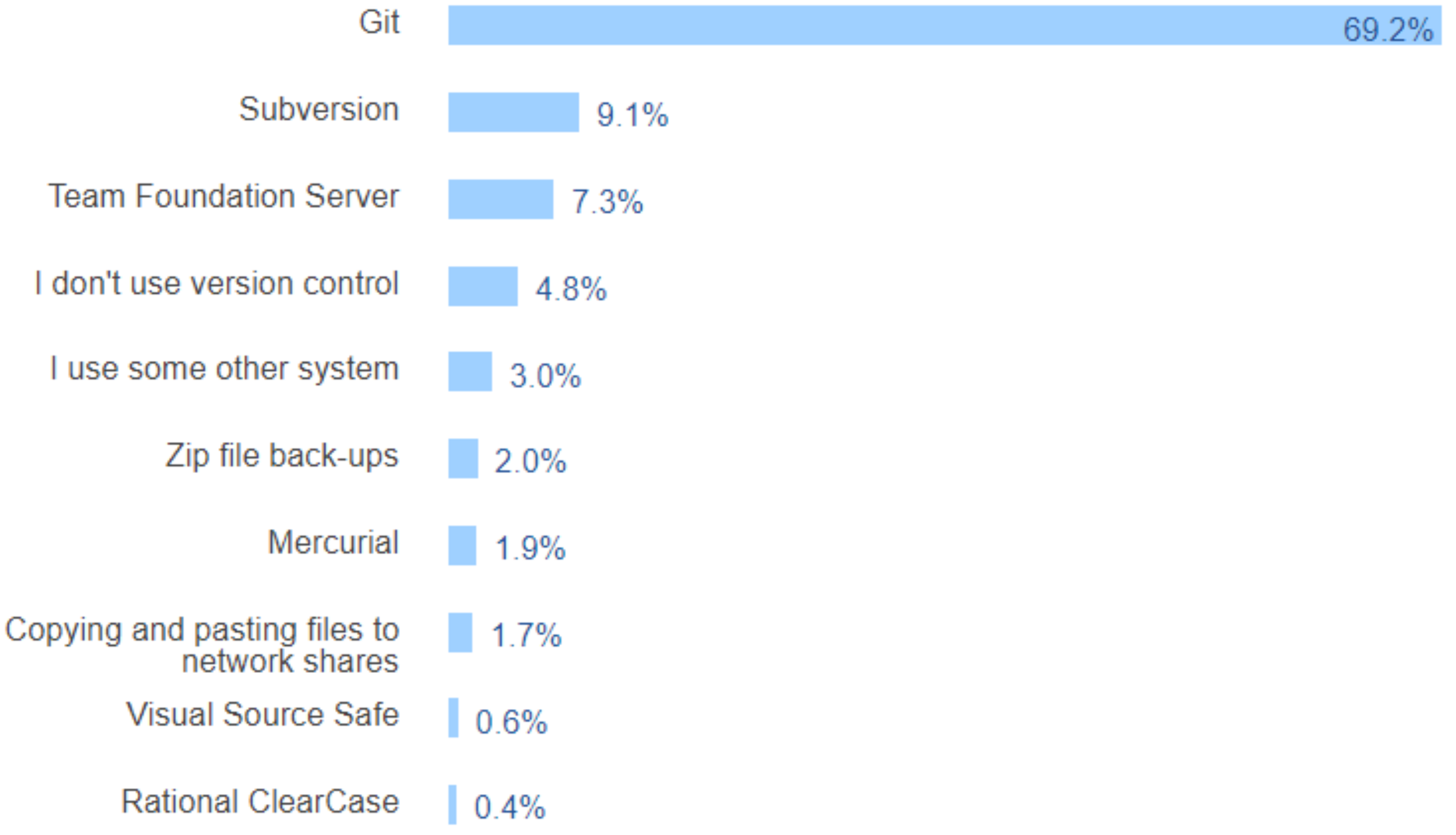
When using a version control system, you have **one copy of each file** and the *version control system tracks the changes* that have occurred over time

What is version control?

A way to manage the evolution of a set of files



The <u>set of files</u> is referred to as a **repository (repo)**


Git
69.2%
Subversion
9.1%
Team Foundation Server
7.3%
I don't use version control
4.8%
I use some other system
3.0%
Zip file back-ups
2.0%
Mercurial
1.9%
Copying and pasting files to network shares
1.7%
Visual Source Safe
0.6%
Rational ClearCase
0.4%

git & GitHub

git

the version control system

~ Track Changes from Microsoft Word….on steroids

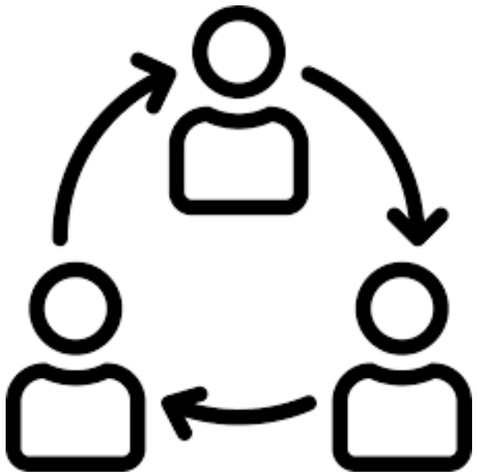
GitHub (or Bitbucket or GitLab) is the home **where your git-based projects live** on the Internet.

~ Dropbox + social media for programmers

Do you even version control?

https://forms.gle/8UeUL2Ux4YtG2CVr8

Why version control with git and GitHub?



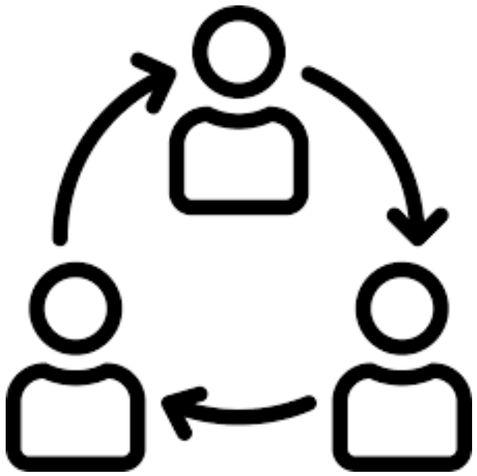
Collaboration

Returning to a safe state

Exposure for your work

Tracking others' work

Collaborate like you do with Google Docs

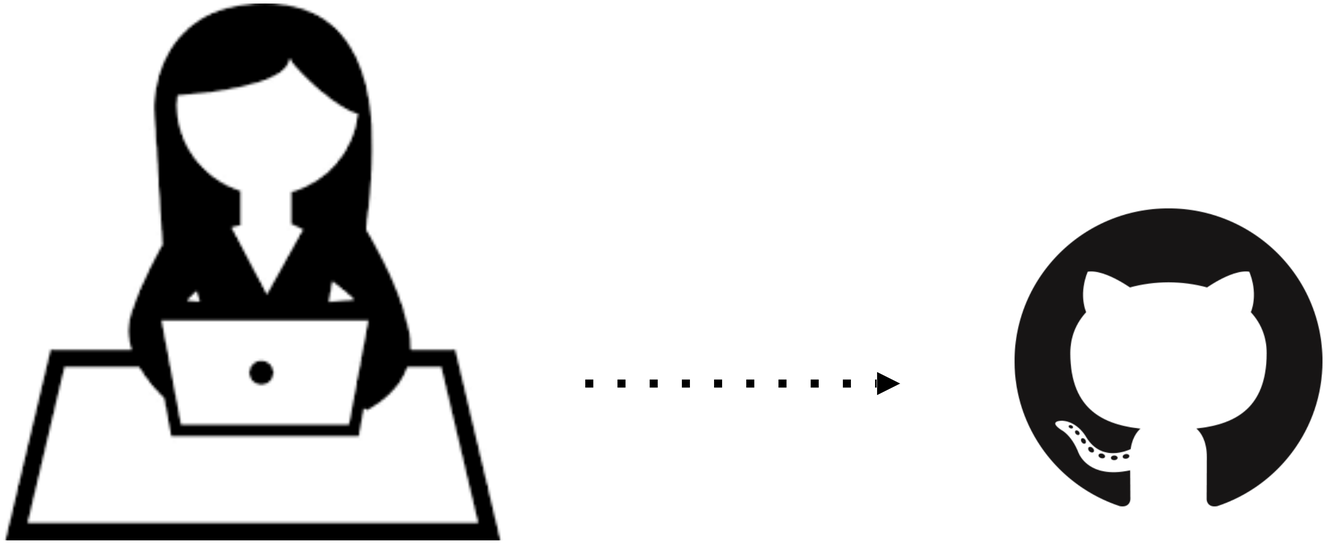


Collaboration

Each person is making changes **locally**
(on their computer)

Make changes locally, while knowing a stable copy exists

Returning to a safe state



You’re free and safe to **try things out locally**. You’ll only send changes to the repo when you’re at a stable point

Your repositories will be visible to others!

Exposure for your work

Your public GitHub repos are your coding social media

And vice versa, you can search for the code you need

For instance, this might come in handy when thinking about class projects

https://github.com/topics/datascience-projects

april-fools

DuckMasterAI / rickroll-bot

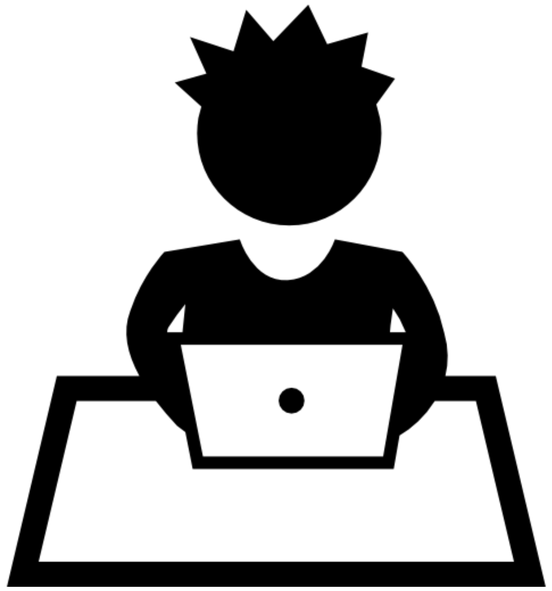
A simple bot to rickroll your friends on Discord!

discord-bot discord-py april-fools rickroll never-gonna-give-you-up never-gonna-let-you-down rick-astley discord-py-bot

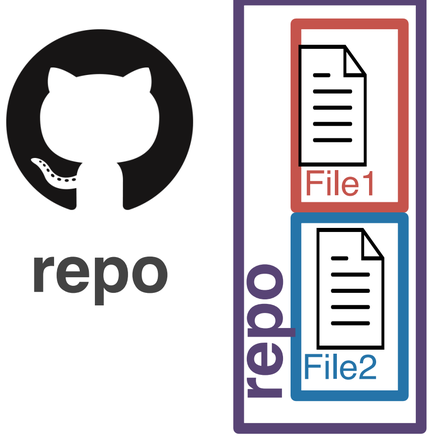
Updated 2 hours ago Python

Keep up with others' work easily

Tracking others' work

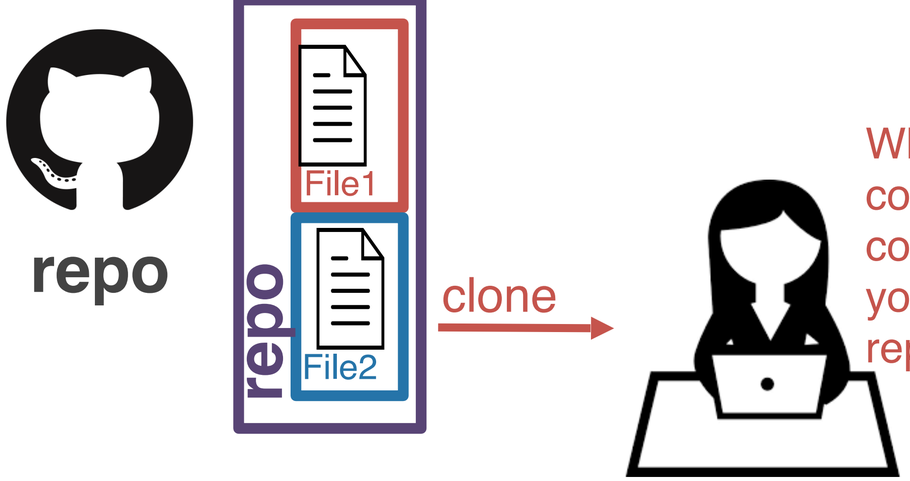


As a social platform, you can see others' work too!

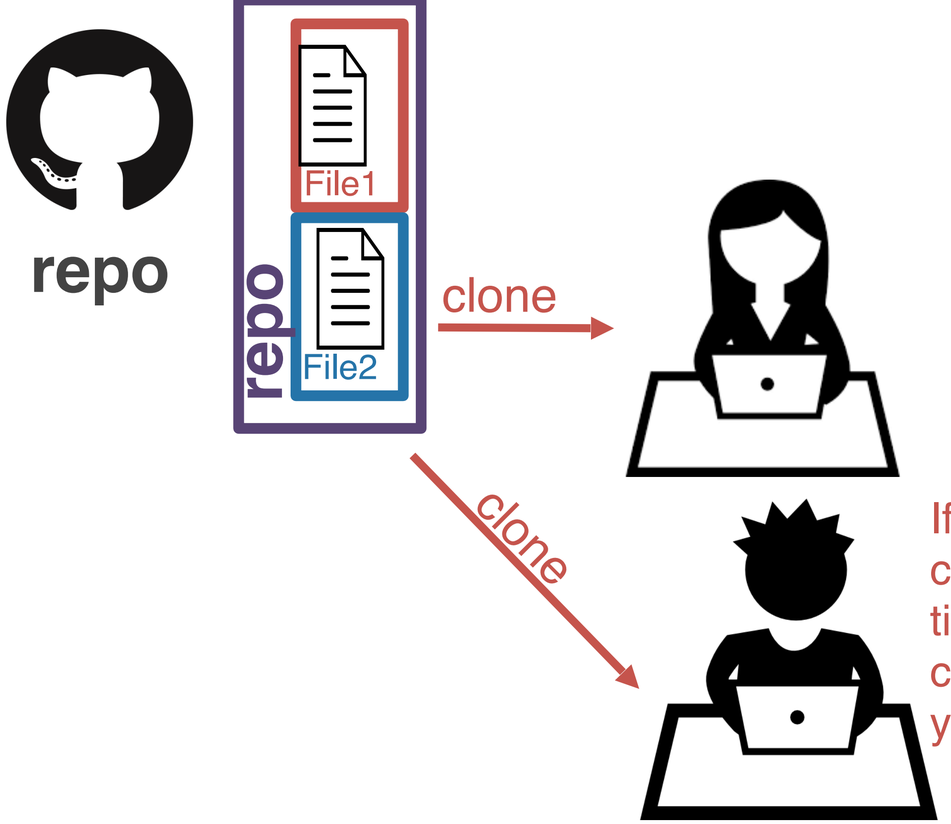


A **GitHub repo** contains all the files and folders for your project.

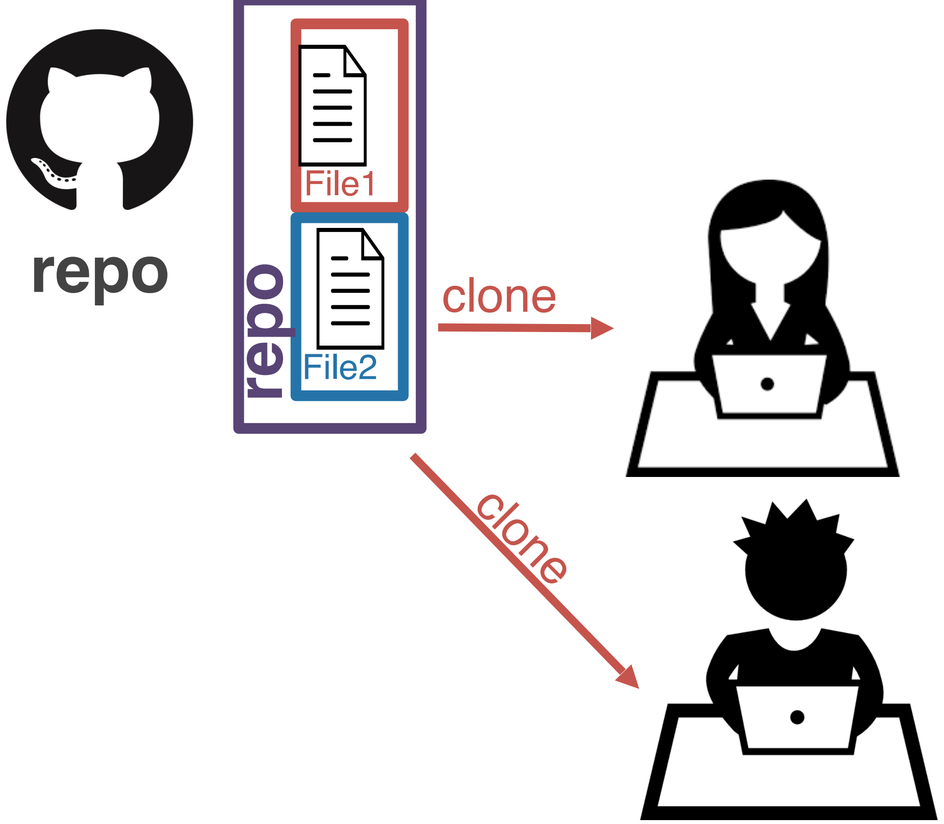
GitHub is a **remote host**. The files are geographically distant from any files on your computer.



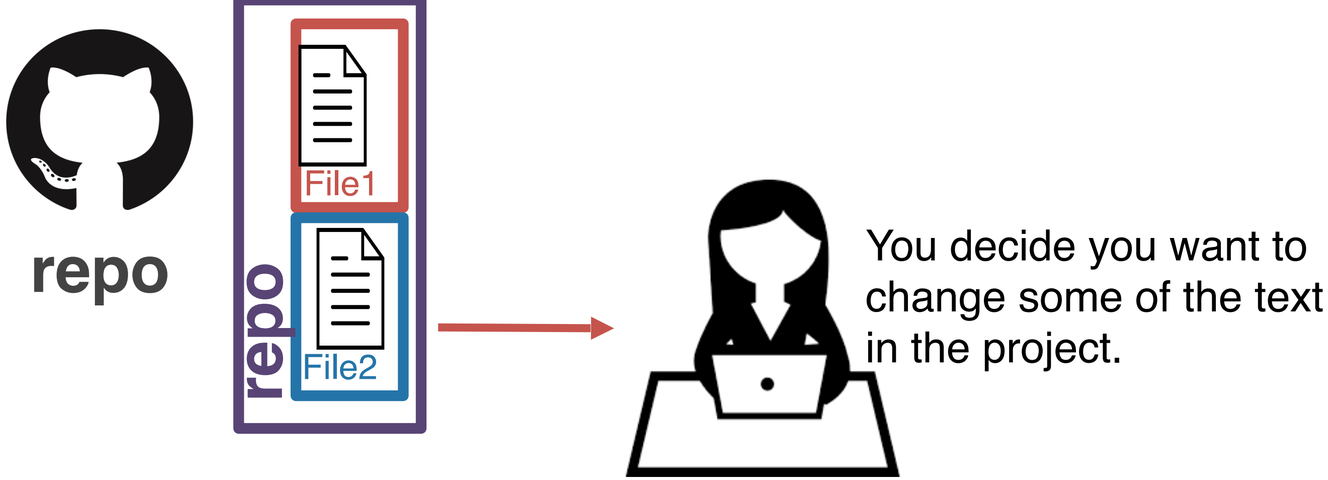
When you first make a copy onto your local computer (read: laptop), you **clone** the repository.

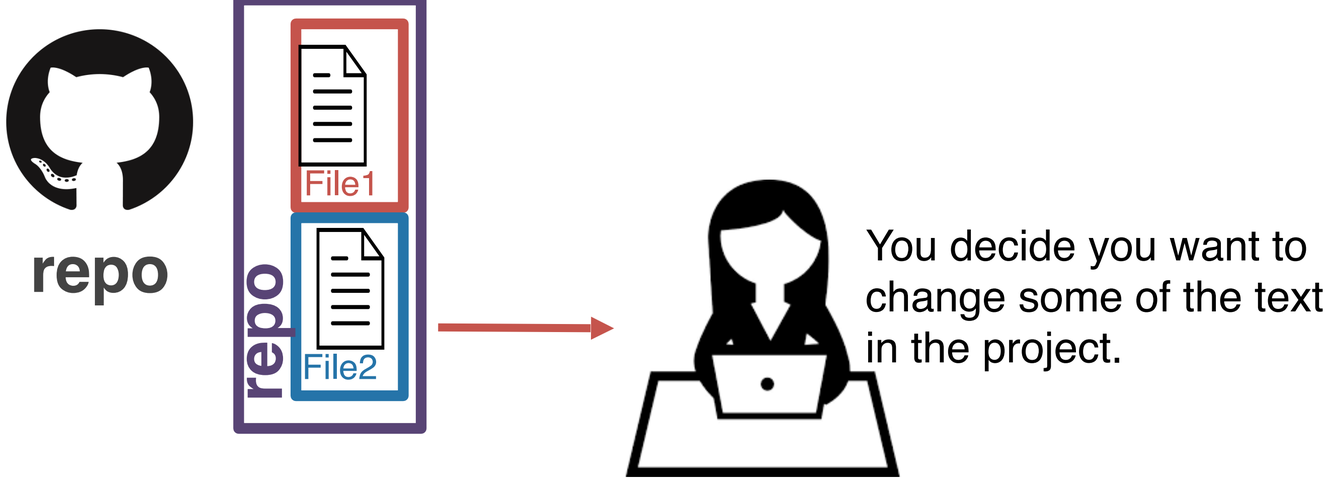


If someone else on your project cloned the repo at the same time, you would have identical copies of the project on each of your computers.



Yay! Everyone can work on the project!


repo
repo
File1
File2
You decide you want to change some of the text in the project.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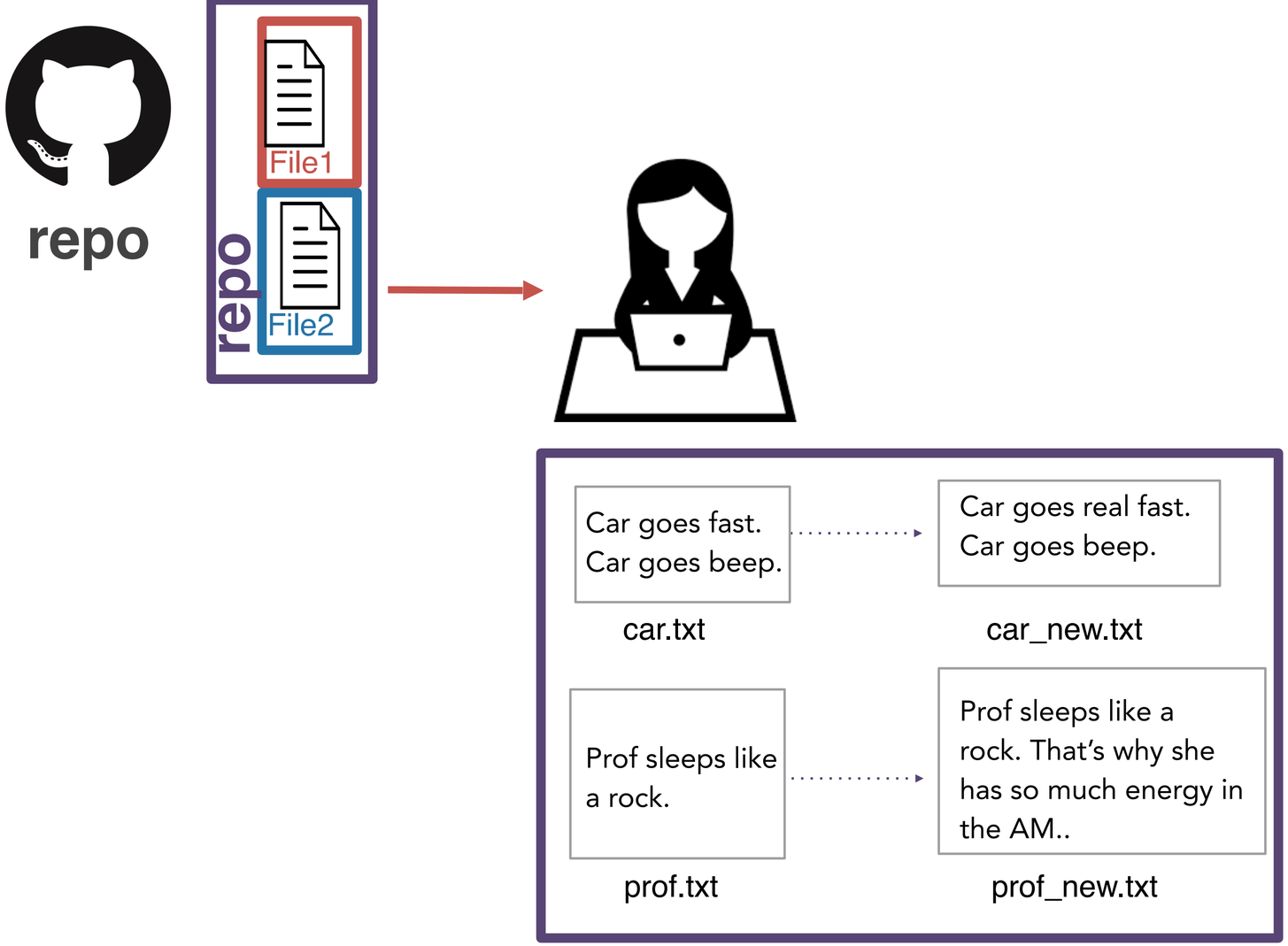
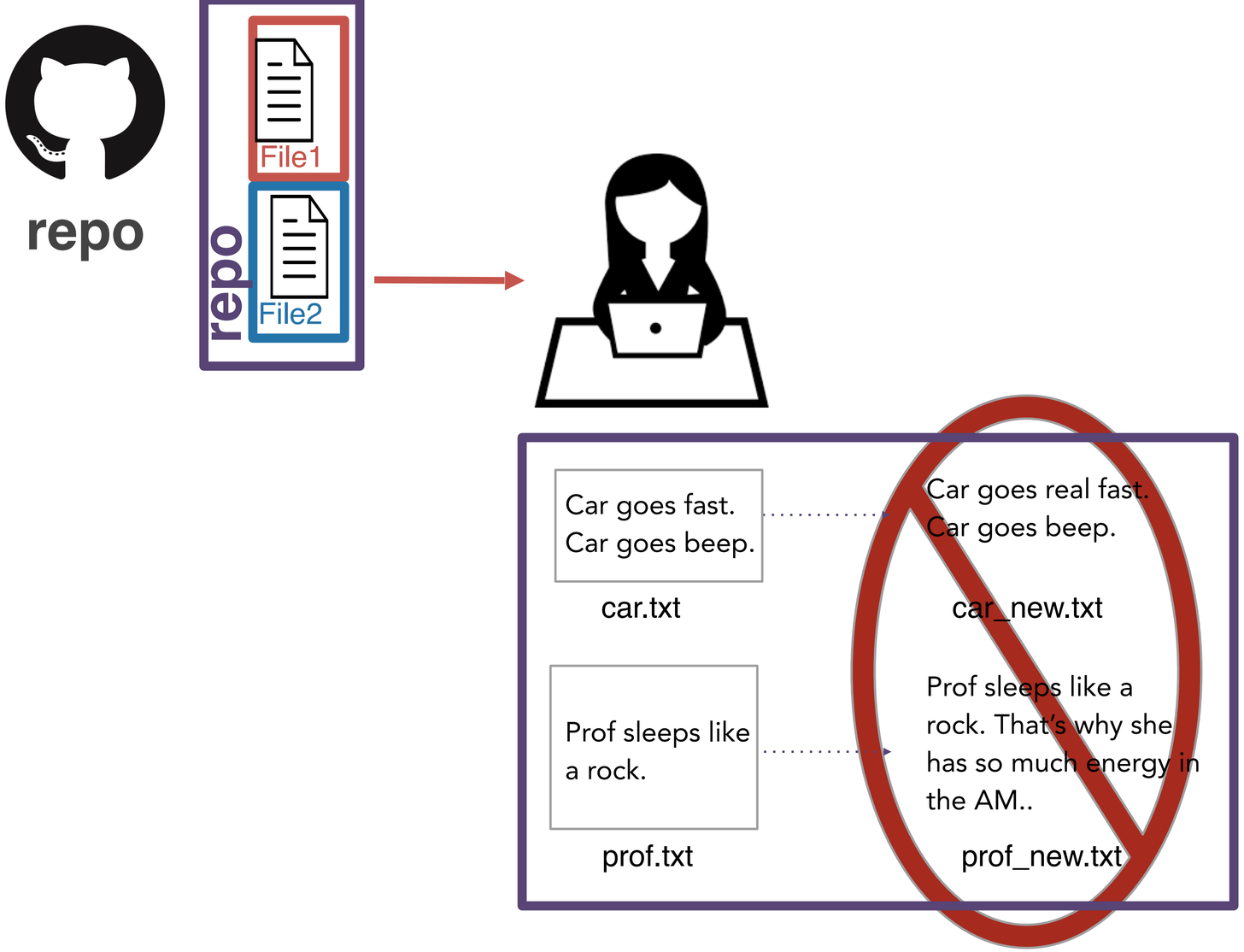
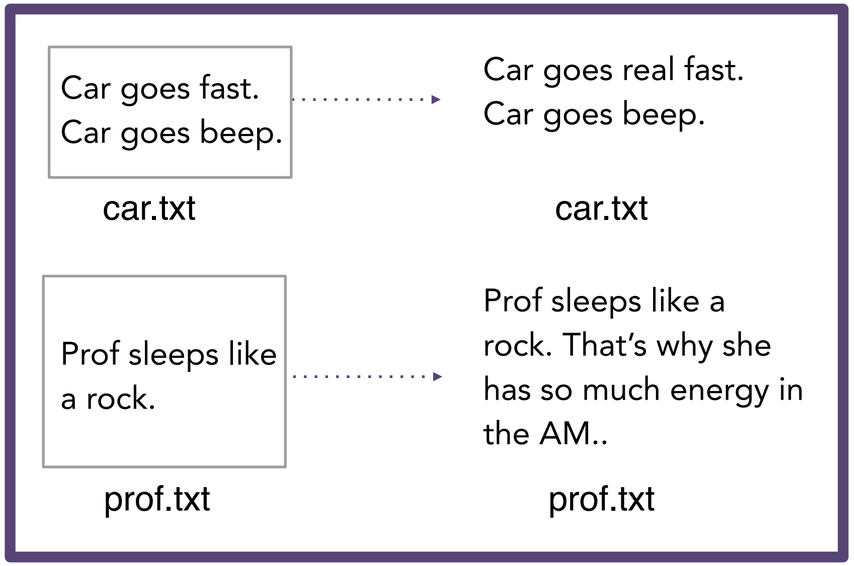


without git...you'd likely rename these files….

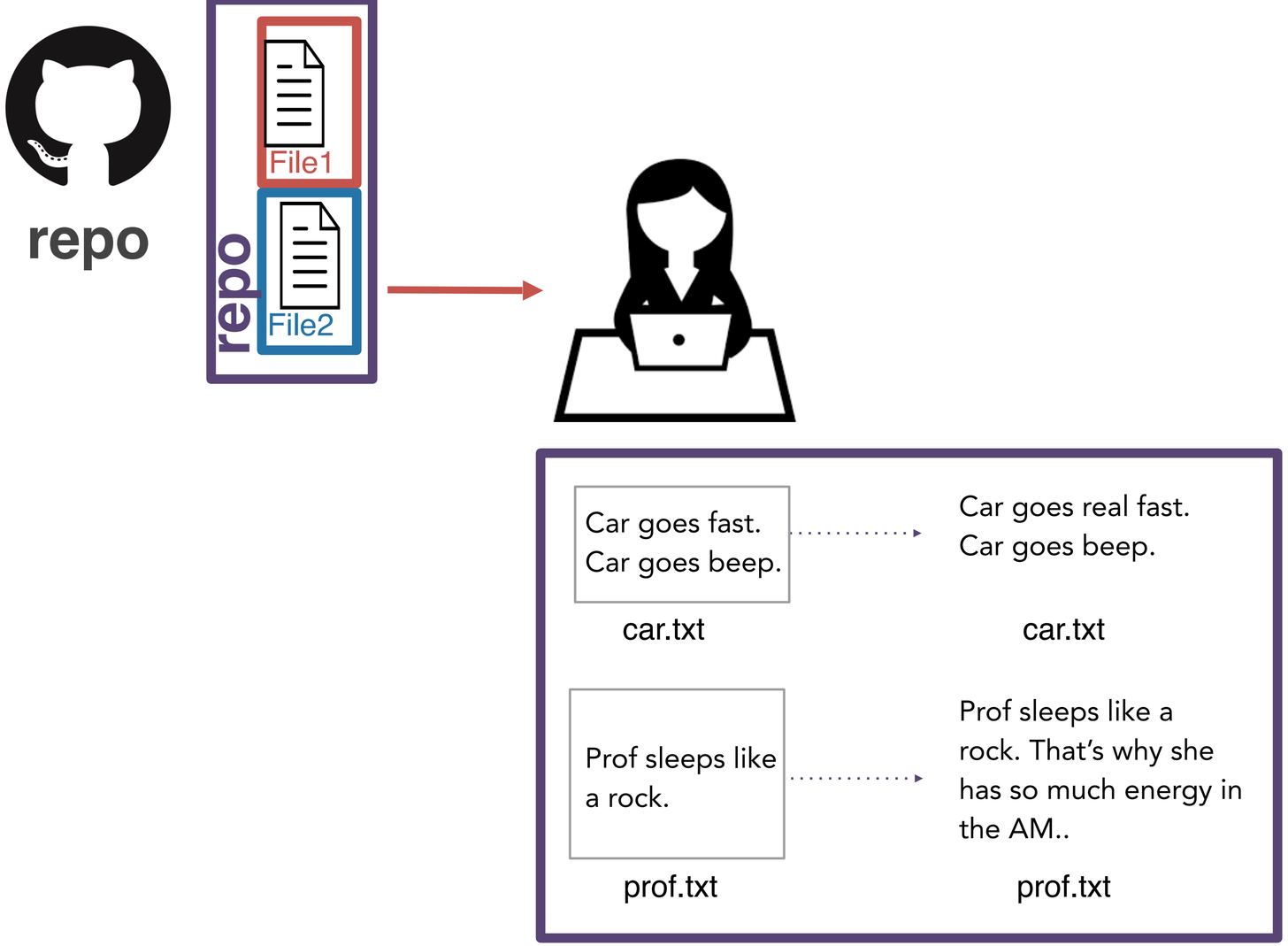


Thank goodness those days are over!

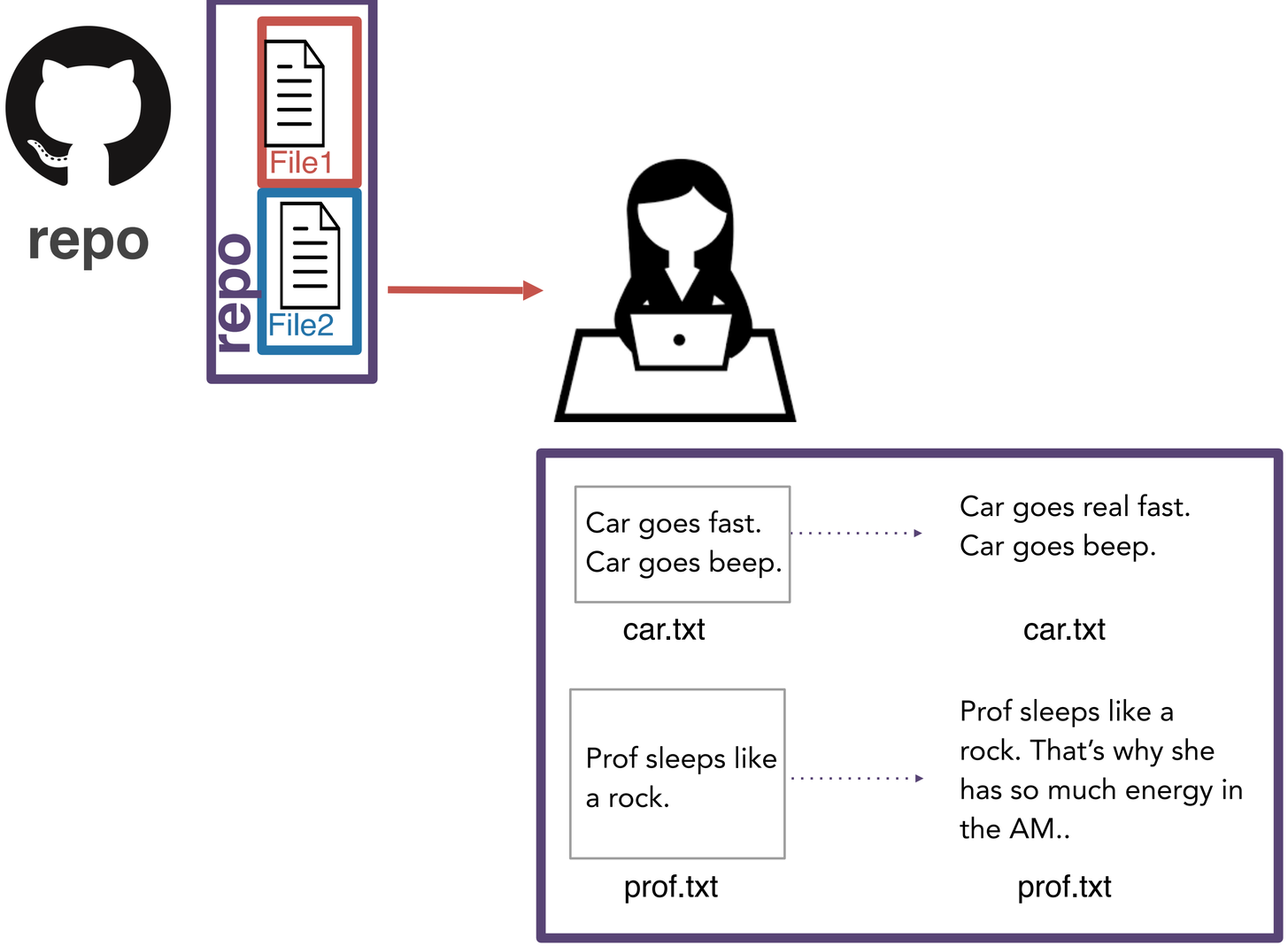
`git add file`	stages specified file (or folder)
`git add .`	stages new and modified files
`git add -u`	stages modified and deleted files
`git add -A`	stages new, modified, and deleted files
`git add *.csv`	stages any files with .csv extension
`git add *`	use with caution: stages everything



Instead, you tell git which files you'd like to keep track of using **add**. This process is called *staging*.

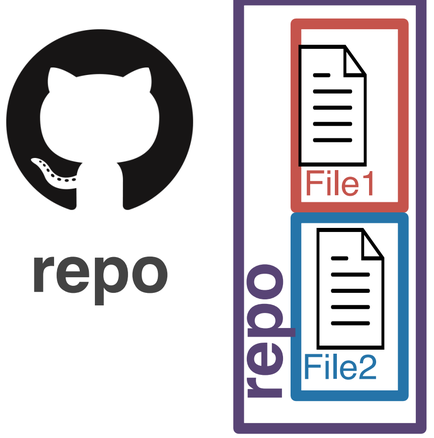


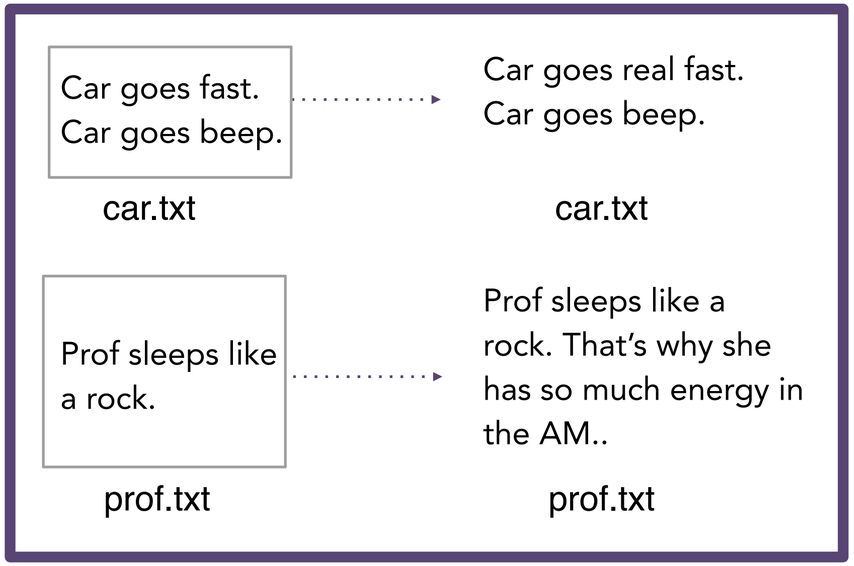
Then, you create a snapshot of your files at this point. This snapshot is called a **commit**.



Then, you create a snapshot of your files at this point. This snapshot is called a **commit**.

A **commit** tracks who, what, and when



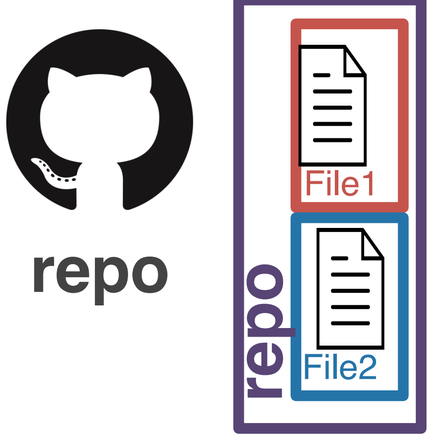


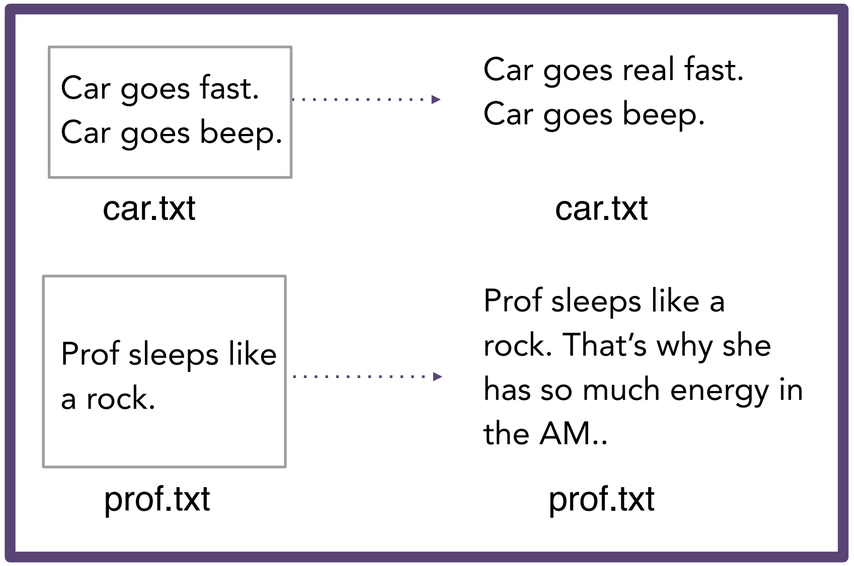
You can make commits more informative by adding a **commit message**.

Example: `git commit -m 'fix typos in car and prof'`

Then, you create a snapshot of your files at this point. This snapshot is called a **commit**.

A **commit** tracks who, what, and when



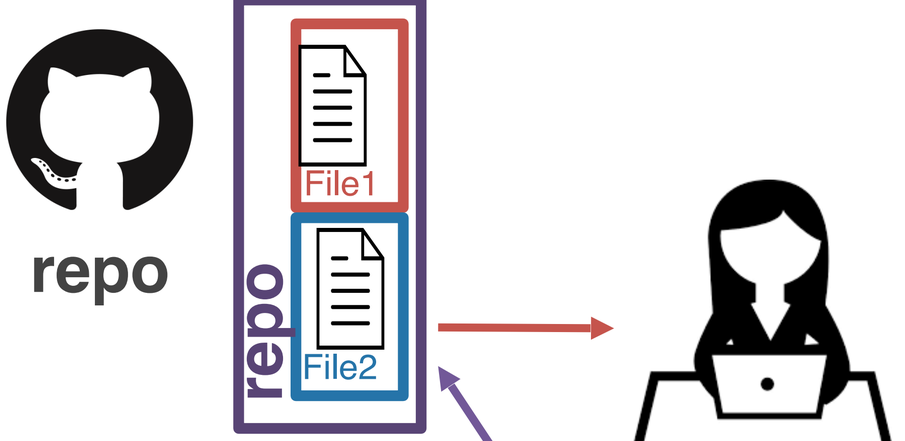


A commit is stored as a **diff**

```
diff --git a/car.txt b/car.txt
index bfd4b76..152d982 100644
--- a/car.txt
+++ b/car.txt
@@ -1,2 +1,2 @@
-Car goes fast
+Car goes real fast
 Car goes beep
diff --git a/prof.txt b/prof.txt
index 1d5eb83..03a85e1 100644
--- a/prof.txt
+++ b/prof.txt
@@ -1 +1,2 @@
 Prof sleeps like a rock
+That's why she has so much energy in the AM..
```

Shannon Ellis
3/28/21 3:28pm

fix typos in car and prof



push

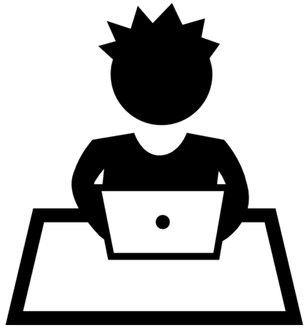
Remember, you're not the only one working on this project though! You want your teammates to have access to these changes! You **push** these changes back to the remote.

Shannon Ellis
3/28/21 3:28pm

fix typos in car and prof

Shannon Ellis
3/28/21 3:28pm

fix typos in car and prof

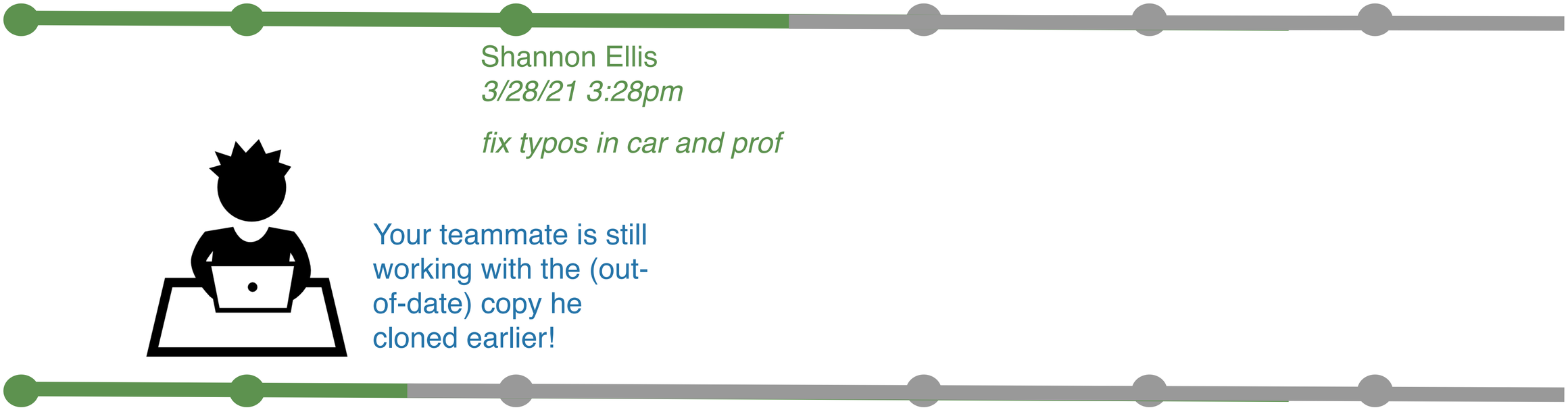


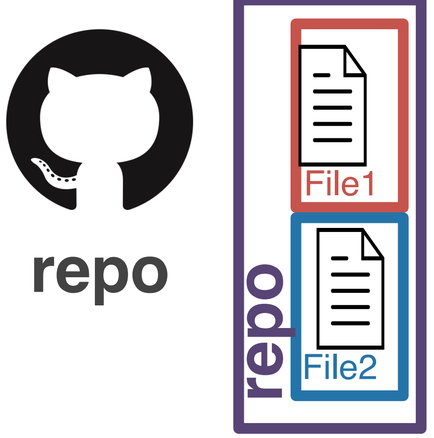
Your teammate is still working with the (out-of-date) copy he cloned earlier!

Jason Fleischer
3/14/21 1:23pm

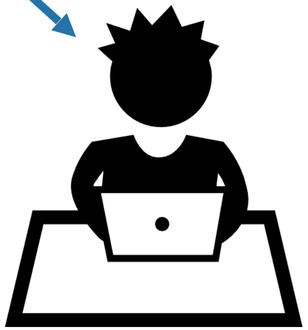
Added file car.txt

To catch up, your teammate will have to **pull** the changes from GitHub (remote)





pull



Your teammate pulls from remote and is now up-to-date!

Shannon Ellis
3/28/21 3:28pm

fix typos in car and prof

The pull transports a **diff** and applies it to your repo

```
diff --git a/car.txt b/car.txt
index bfd4b76..152d982 100644
--- a/car.txt
+++ b/car.txt
@@ -1,2 +1,2 @@
-Car goes fast
+Car goes real fast
 Car goes beep
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 Prof sleeps like a rock
+That's why she has so much energy in the AM...
```
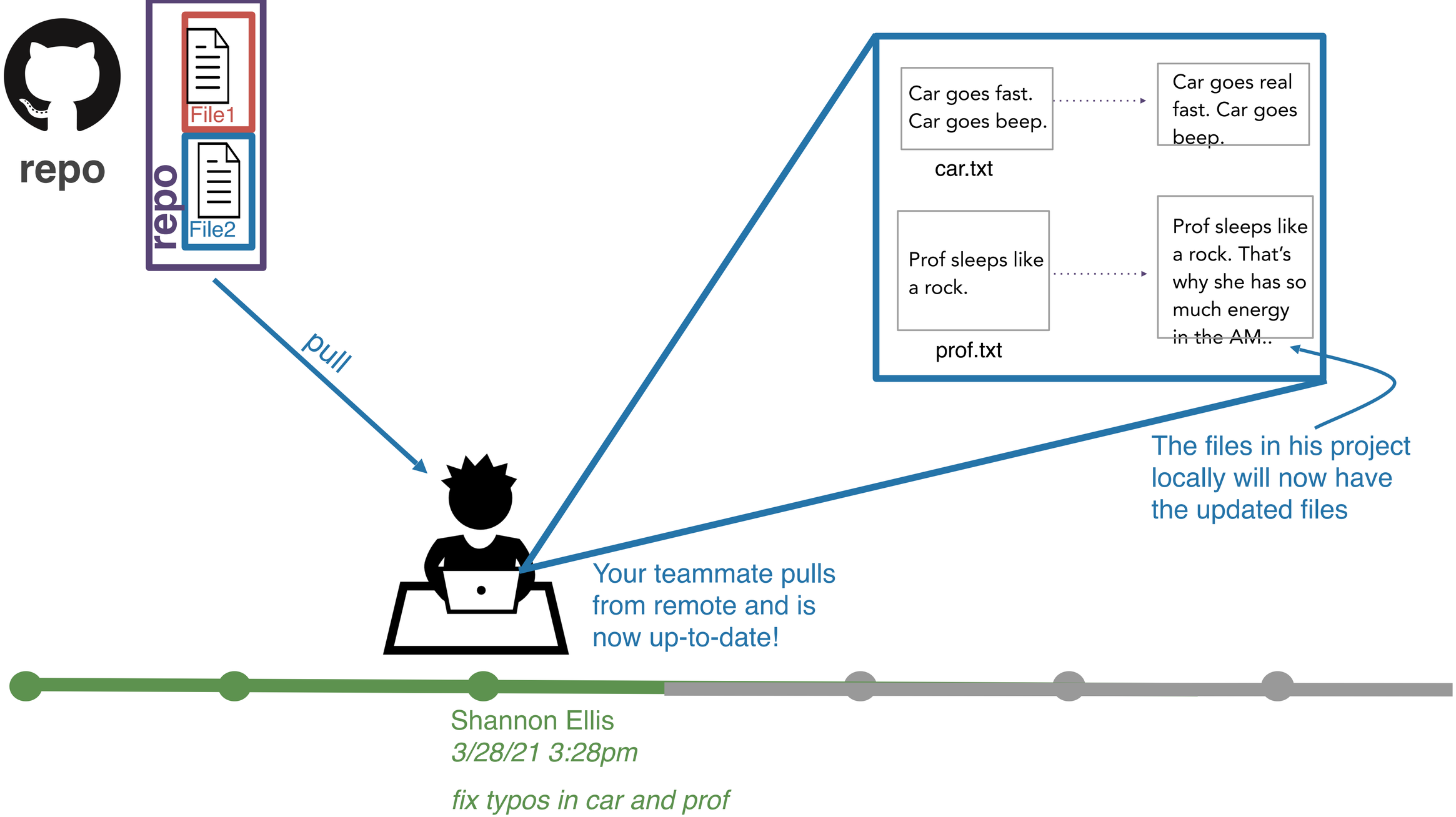
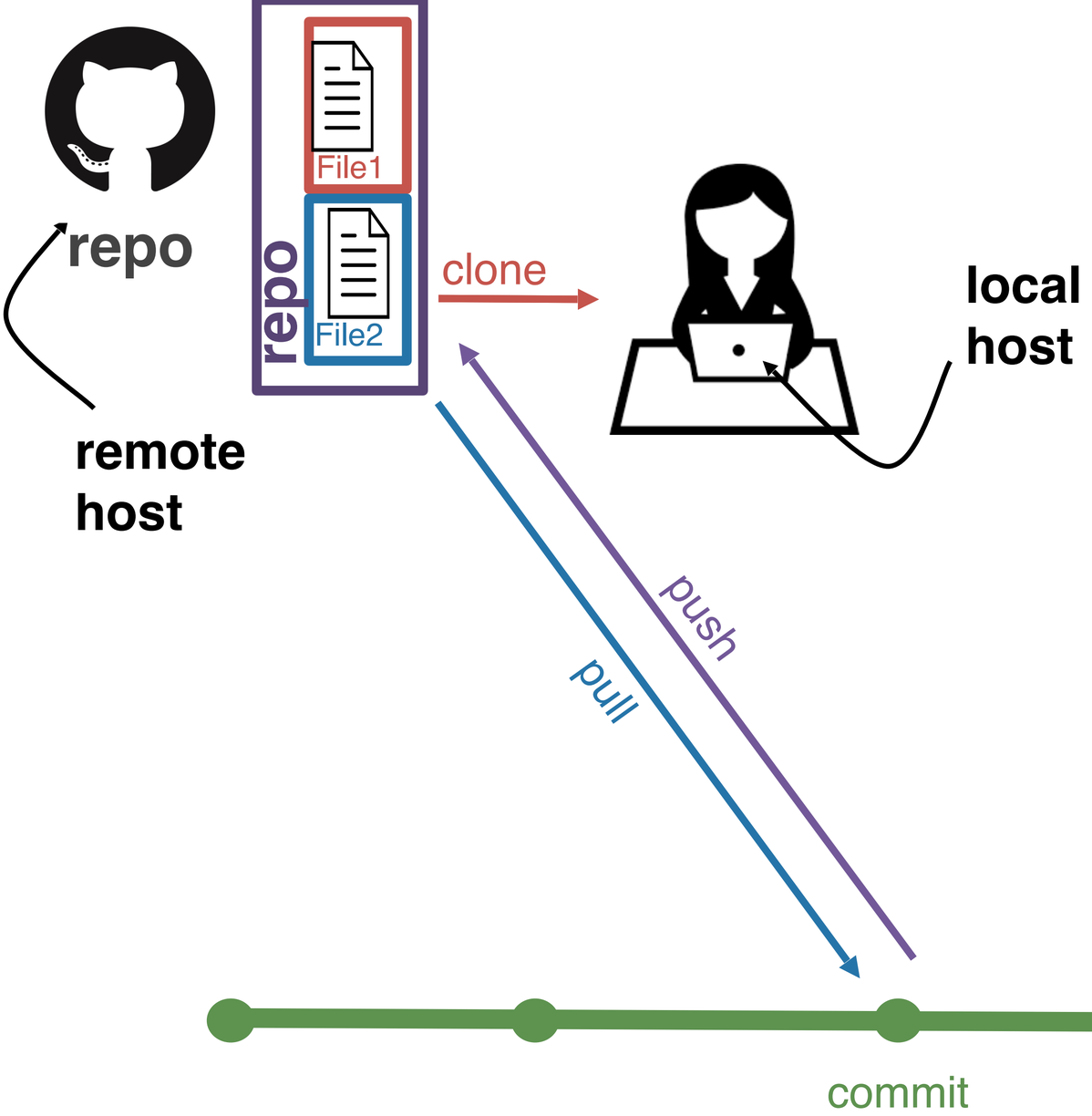


repo
repo
File1
File2
pull
Car goes fast. Car goes beep.
Car goes real fast. Car goes beep.
car.txt
Prof sleeps like a rock.
Prof sleeps like a rock. That's why she has so much energy in the AM..
prof.txt
The files in his project locally will now have the updated files
Your teammate pulls from remote and is now up-to-date!
Shannon Ellis
3/28/21 3:28pm
fix typos in car and prof



Let’s recap real quick!

repo - set of files and folders for a project
remote - where the repo lives
clone - get the repo from the remote for the first time
add - specify which files you want to stage (add to repo)
commit - snapshot of your files at a point in time
pull - get new commits to the repo from the remote
push - send your new commits to the remote

Basic version control

https://forms.gle/wHA2GSyuycFre5qr6

What version control looks like

```
$ git clone https://www.github.com/username/repo.git
$ git pull

[… edit some files, make changes …]

$ git add -A
$ git commit -m “informative commit message”
$ git push
```

Git on command line

Git exercise #1

1. Login to datahub.ucsd.edu
2. Open a terminal
3. In the terminal type:

```
git config --global user.email “you@ucsd.edu”
git config --global user.name “Your Name”

git clone https://github.com/COGS108/Lectures-So26
```

4. Ask yourself, how will the command look different when I want to update my lecture repo with the new slides next week. Try that command out!

Git exercise #2 - updating a repo

1. Change directory to the new repo Lectures-Sp26
2. Figure out if what (if anything) is different from the last commit, type:
   `git status`
3. Create a brand new file
   `touch 007-new-movie.mov`
4. Figure out if what (if anything) is different from the last commit, type:
   `git status`
5. Stage your changes
   `git add 007-new-movie.mov`
6. Commit your changes
   `git commit -m “unreleased James Bond movie”`
7. Question: What happens if you push to Github? Why?
8. Undo these unwanted changes to prevent problems in the future:
   `git reset —hard HEAD~1`